

Summary Report: Candidate literary paper to be used

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Contents

Paper 1: On the four types of weight functions for spatial contiguity matrix

Summary

The studies of spatial complexity have resulted in new understanding of the distance-decay functions of geography. The weight functions of spatial autocorrelation should be consistent with the functions of spatial distribution and interaction. This paper is on “spatial autocorrelation analysis” of spatial autocorrelation. The 1-dimension spatial autocorrelation function (ACF) and partial autocorrelation function (PACF) of space series are employed to analyze four kinds of weight function in common use for the 2-dimensional spatial autocorrelation model. The aim of this study is at how to select a proper weight function to construct a spatial contiguity matrix for spatial analysis. The scopes of application of different weight functions are defined in terms of the characters of their ACFs and PACFs. Chen 2012

Metadata

1. **Link:** <https://link.springer.com/article/10.1007/s12076-011-0076-6>
2. **Journal Name:** Letters in Spatial and Resource Sciences
3. **Rank:** Q2

Conclusions

- **Objective:** To provide an objective criterion for selecting spatial weight functions based on the consistency between spatial distribution, interaction, and scale.
- **Methodology:** The author uses 1-dimensional Autocorrelation Functions (ACF) and Partial Autocorrelation Functions (PACF) to analyze the behavior and “spatial memory” of four common weight functions used in 2-dimensional spatial analysis.
- **Key Findings:**
 - Weight functions must match the underlying geographical process (e.g., power-law vs. exponential distributions).
 - The choice of function is critical at “critical states” (the transition between chaos and order), where different functions yield significantly different analytical results.
- **Conclusion:**
 - **Power functions** are best for large-scale complex systems.
 - **Exponential functions** suit medium-to-small scale simple systems.
 - **Step functions** are ideal for local-level analysis and negative correlation patterns.

Paper 2: Autoregressive Spatial Smoothing and Temporal Spline Smoothing for Mapping Rates

Summary

This article proposes generalized additive mixed models for the analysis of geographic and temporal variability of mortality rates. This class of models accommodates random spatial effects and fixed and random temporal components. Spatiotemporal models that use autoregressive local smoothing across the spatial dimension and B-spline smoothing over the temporal dimension are developed. The objective is the identification of temporal trends and the production of a series of smoothed maps from which spatial patterns of mortality risks can be monitored over time. Regions with consistently high rate estimates may be followed for further investigation. The methodology is illustrated by analysis of British Columbia infant mortality data. MacNab and Dean 2001 Chen 2012

Metadata

1. **Link:** <https://doi.org/10.1111/j.0006-341X.2001.00949.x>
2. **Journal Name:** Biometrics Journal of the international Biometric society
3. **Rank:** Q1

Paper 3: On confidence intervals for generalized additive models based on penalized regression splines

Summary

Generalized additive models represented using low rank penalized regression splines, estimated by penalized likelihood maximisation and with smoothness selected by generalized cross validation or similar criteria, provide a computationally efficient general framework for practical smooth modelling. Various authors have proposed approximate Bayesian interval estimates for such models, based on extensions of the work of Wahba, G. (1983)[Bayesian confidence intervals for the cross validated smoothing spline. J. R. Statist. Soc. B45, 133–150] and Silverman, B.W. (1985)[Some aspects of the spline smoothing approach to nonparametric regression curve fitting. J. R. Statist. Soc. B47, 1–52] on smoothing spline models of Gaussian data, but testing of such intervals has been rather limited and there is little supporting theory for the approximations used in the generalized case. This paper aims to improve this situation by providing simulation tests and obtaining asymptotic results supporting the approximations employed for the generalized case. The simulation results suggest that while across-the-model performance is good, component-wise coverage probabilities are not as reliable. Since this is likely to result from the neglect of smoothing parameter variability, a simple and efficient simulation method is proposed to account for smoothing parameter uncertainty: this is demonstrated to substantially improve the performance of component-wise intervals. Wood 2006

Metadata

1. **Link:** <https://doi.org/10.1111/j.1467-842X.2006.00450.x>
2. **Journal Name:** Australian Statistical Publishing Association
3. **Rank:** Q3

Paper 4: GAMs with integrated model selection using penalized regression splines and applications to environmental modelling

Summary

Generalized additive models (GAMs) have been popularized by the work of Hastie and Tibshirani (Generalized Additive Models (1990)) and the availability of user friendly gam software in Spls. However, whilst it is flexible and efficient, the gam framework based on backfitting with linear smoothers presents some difficulties when it comes to model selection and inference. On the other hand, the mathematically elegant work of Wahba (Spline Models for

Observational Data (1990)) and co-workers on Generalized Spline Smoothing (GSS) provides a rigorous framework for model selection (SIAM J. Sci. Statist. Comput. 12 (1991) 383) and inference with GAMs constructed from smoothing splines: but unfortunately these models are computationally very expensive with operations counts that are of cubic order in the number of data. A ‘middle way’ between these approaches is to construct GAMs using penalized regression splines (e.g. Marx and Eilers, Comput. Statist. Data Anal. (1998)). In this paper, we develop this idea further and show how GAMs constructed using penalized regression splines can be used to get most of the practical benefits of GSS models, including well founded model selection and multi-dimensional smooth terms, with the ease of use and low computational cost of backfit GAMs. Inference with the resulting methods also requires slightly fewer approximations than are employed in the GAM modelling software provided in Splus. This paper presents the basic mathematical and numerical approach to GAMs implemented in the R package mgcv, and includes two environmental examples using the methods as implemented in the package. Wood and Augustin 2002

Metadata

1. **Link:** [https://doi.org/10.1016/S0304-3800\(02\)00193-X](https://doi.org/10.1016/S0304-3800(02)00193-X)
2. **Journal Name:** Ecological modelling
3. **Rank:** Q1

Paper 5: Thin Plate Regression Splines

Summary

I discuss the production of low rank smoothers for $d \geq 1$ dimensional data, which can be fitted by regression or penalized regression methods. The smoothers are constructed by a simple transformation and truncation of the basis that arises from the solution of the thin plate spline smoothing problem and are optimal in the sense that the truncation is designed to result in the minimum possible perturbation of the thin plate spline smoothing problem given the dimension of the basis used to construct the smoother. By making use of Lanczos iteration the basis change and truncation are computationally efficient. The smoothers allow the use of approximate thin plate spline models with large data sets, avoid the problems that are associated with ‘knot placement’ that usually complicate modelling with regression splines or penalized regression splines, provide a sensible way of modelling interaction terms in generalized additive models, provide low rank approximations to generalized smoothing spline models, appropriate for use with large data sets, provide a means for incorporating smooth functions of more than one variable into non-linear models and improve the computational efficiency of penalized likelihood models incorporating thin plate splines. Given that the approach produces spline-like models with a sparse basis, it also provides a natural way of incorporating unpenalized spline-like terms in linear and generalized linear models, and these can be treated just like any other model terms from the point of view of model selection, inference and diagnostics. Wood 2003

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Metadata

1. **Link:** <https://doi.org/10.1111/1467-9868.00374>
2. **Journal Name:** Journal of the Royal Statistical Society Series B: Statistical Methodology
3. **Rank:** Q1

Paper 6: Selecting the number of knots for penalized splines

Summary

Penalized splines, or P-splines, are regression splines fit by least-squares with a roughness penalty. P-splines have much in common with smoothing splines, but the type of penalty used with a P-spline is somewhat more general than for a smoothing spline. Also, the number and location of the knots of a P-spline is not fixed as with a smoothing spline. Generally, the knots of a P-spline are at fixed quantiles of the independent variable and the only tuning parameters to choose are the number of knots and the penalty parameter. In this article, the effects of the number of knots on

the performance of P-splines are studied. Two algorithms are proposed for the automatic selection of the number of knots. The myopic algorithm stops when no improvement in the generalized cross-validation statistic (GCV) is noticed with the last increase in the number of knots. The full search examines all candidates in a fixed sequence of possible numbers of knots and chooses the candidate that minimizes GCV. The myopic algorithm works well in many cases but can stop prematurely. The full-search algorithm worked well in all examples examined. A Demmler–Reinsch type diagonalization for computing univariate and additive P-splines is described. The Demmler–Reinsch basis is not effective for smoothing splines because smoothing splines have too many knots. For P-splines, however, the Demmler–Reinsch basis is very useful for super-fast generalized cross-validation. Ruppert 2002

Metadata

1. **Link:** <https://doi.org/10.1198/106186002853>
2. **Journal Name:** Journal of computational and graphical statistics
3. **Rank:** Q1

Paper 7: A review of performance criteria to validate simulation models

Summary

This study reviews performance criteria adequate to validate simulation models through the comparison of two quantitative data sets, concerning historical and simulated data. The criteria reviewed were organized according to its characteristics into the groups: error-based measures, information theory measures, information criteria, parametric tests, non-parametric tests, distance-based measures and combined measures. Each criterion is reviewed through its mathematic definition, its applications in literature and the identification of its advantages and drawbacks. The features assessed by each criterion are identified and discussed. This study provides a concise outline over the criteria reviewed, which can be used as a guide to help developers of simulation models into the decision on the most appropriate criteria to validate their models. Hora and Campos 2015

Metadata

1. **Link:** <https://doi.org/10.1111/exsy.12111>
2. **Journal Name:** Expert Systems
3. **Rank:** Q1

Spatial smoothing in Bayesian models: a comparison of weights matrix specifications and their impact on inference

Summary

Background: When analysing spatial data, it is important to account for spatial autocorrelation. In Bayesian statistics, spatial autocorrelation is commonly modelled by the intrinsic conditional autoregressive prior distribution. At the heart of this model is a spatial weights matrix which controls the behaviour and degree of spatial smoothing. The purpose of this study is to review the main specifications of the spatial weights matrix found in the literature, and together with some new and less common specifications, compare the effect that they have on smoothing and model performance.

Methods: The popular BYM model is described, and a simple solution for addressing the identifiability issue among the spatial random effects is provided. Seventeen different definitions of the spatial weights matrix are defined, which are classified into four classes: adjacency-based weights, and weights based on geographic distance, distance between covariate values, and a hybrid of geographic and covariate distances. These last two definitions embody the main novelty of this research. Three synthetic data sets are generated, each representing a different underlying spatial structure. These data sets together with a real spatial data set from the literature are analysed using the models. The models are evaluated using the deviance information criterion and Moran’s I statistic. Results: The deviance information criterion indicated that the model which uses binary, first-order adjacency weights to perform spatial smoothing is generally an optimal choice for achieving a good model fit. Distance-based weights also generally

perform quite well and offer similar parameter interpretations. The less commonly explored options for performing spatial smoothing generally provided a worse model fit than models with more traditional approaches to smoothing, but usually outperformed the benchmark model which did not conduct spatial smoothing.

Conclusions: The specification of the spatial weights matrix can have a colossal impact on model fit and parameter estimation. The results provide some evidence that a smaller number of neighbours used in defining the spatial weights matrix yields a better model fit, and may provide a more accurate representation of the underlying spatial random field. The commonly used binary, first-order adjacency weights still appear to be a good choice for implementing spatial smoothing. Duncan, White, and Mengersen 2017

Metadata

1. **Link:**
2. **Journal Name:** International journal of health geographics
3. **Rank:** Q1

Template

Summary

Chen 2012

Metadata

1. **Link:**
2. **Journal Name:**
3. **Rank:**

Conclusions

References

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